Improving the availability of waterbird data for environmental emergency planning and response

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Background

Introduction

The mandate of the Canadian Wildlife Service (CWS) includes the management of all Migratory Birds and Species at Risk under its jurisdiction during environmental emergencies (an uncontrolled or unexpected incident involving the release of a polluting substance into the environment that may result in an immediate or long-term harmful effect on the environment) and Pollution Incidents (an incident involving the release or deposit of a substance that is harmful to wildlife). If such an incident or emergency results or may result in an immediate and/or long-term harmful effect on the life or health of wildlife and/or their habitat, it may be deemed a Wildlife Emergency. Determining whether any specific incident should be deemed a Wildlife Emergency depends on factors including the scope and severity of the incident (e.g., numbers of animals or area of habitat impacted), the potential for impacts to Species at Risk.

Wildlife Response Plans to guide response in the case of an environmental emergency are required for construction, operation or decommissioning activities that may impact the environment, and may also be required for some projects as part of an environmental assessment process. It is recommended that strategic response plans that describe the likely activities required to respond to an emergency be developed in advance for specific higher-risk activities and/or areas. However, in order to develop such plans, managers need to be able to identify areas of higher risk or sensitivity to wildlife or their habitats, as well as the species at risk of being impacted.

Similarly, when responding to an environmental emergency, CWS requires information such as which habitats are particularly sensitive or of concern, the numbers of Migratory Birds and Species at Risk likely present in that area at that time of year to assess potential impacts on wildlife populations and effectively guide mitigation measures to the highest priority locations. An initial wildlife impact assessment needs to be conducted as early in the incident response as possible (within the first 24 hours, according to guidelines; ECCC 2022). Thus, it is critical for CWS and other wildlife managers to have good baseline information on the distribution and abundance of Wildlife in known or potential high-risk areas and sensitive habitats, and to have this information needs to be readily available and compiled into a single dataset.

Although pollution that is discharged into a terrestrial landscape can potentially impact birds, mammals, reptiles, and amphibians, the spread is more limited than in open water (e.g., large bodies of freshwater) and aquatic (e.g., creeks, rivers, marshes, fens and bogs) environments. As a result, waterfowl, shorebirds, grebes, terns/gulls and other species with similar life history requirements (hereafter waterbirds) tend to be at higher risk for exposure to pollutants in environmental emergencies. Having a strong baseline understanding of which species may use different aquatic habitat types will therefore aid in assessing risks if an environmental emergency occurs. Species that tend to congregate in large numbers, for example when breeding (e.g., colonial breeding terns) or staging and moulting (e.g., waterfowl, shorebirds), are of particularly high concern when planning for wildlife emergencies, as the discharge of even a low volume of a contaminant can impact very large numbers of individuals. Therefore, the identification of habitats and specific locations where

these congregations occur is also of high priority when planning strategic response plans for environmental emergencies.

Within the Prairie Region of CWS, environmental emergency preparedness is advanced within the prairie ecosystem regions, which have historically been the areas most likely to be impacted by wildlife emergencies. However, the boreal region is increasingly at risk of environmental emergencies from industrial developments (e.g., mines) and the associated transportation pressures (e.g., transport of goods by road or rail). While collection and interpretation of baseline data on landbirds is reasonably advanced within this region, such information is largely lacking for waterbirds.

Goal & Objectives

To support planning and decision-making, the report comprises several analytical objectives that will increase our baseline understanding of waterbird distribution within the region: 1. Compile all available data from surveys in or near wetland habitats in AB, SK and MB; 2. Determine appropriate methods to harmonize the data from different survey methods into a common data format; 3. Import these data into WildTrax, an open data platform widely used by ECCC and other organizations; 4. Determine how to remove upland surveys; 5. Produce a single dataset, comprising only surveys of wetland habitats, available for analysis; 6. Produce a short report summarizing the wetland habitat associations for each waterbird species; 7. Determine the extent of unknown species identifications in recordings, and recommend an approach to reduce the number of such identifications in the future

Deliverables

These objectives will result in the following deliverables: 1. A report describing methods used to harmonize data; 2. List of WildTrax projects added, including whether each has additional habitat data available; 3. A report describing methods used to remove surveys in upland habitat types from the dataset; 4. A harmonized dataset, comprising only surveys of wetland habitats, that can be used for analysis; 5. A report summarizing basic wetland habitat associations for each waterbird species (for species which are detected in sufficiently large numbers to support analysis); 6. A report determining the extent of unknown species identifications in recordings (e.g., unknown shorebird, unknown duck), and recommending an approach to reduce the number of such identifications in the future.

All code for this report and the supporting analyses is available in a GitHub repository at https://github.com/borealbirds/Waterbirds.

Data Compilation & Harmonization

The following section describes methods and results for data compilation and harmonization (Deliverables 1 and 2).

Datasets

A comprehensive survey of surveys for shorebirds, waterbirds, and marshbirds in or neat wetland habitats in AB, SK, and MB was undertaken. We worked with external agencies to identify potential datasets that are not yet available in WildTrax (e.g., Ducks Unlimited Canada, Audubon Canada, Birds Canada, marsh monitoring program, CWS). We also included surveys from other regions of Canada to inform subsequent analyses. We considered the following data types in our survey: - Autonomous recording unit (ARU) surveys - Point count survey - Point count surveys with call playbacks - Aerial surveys - Plot search surveys

Ten datasets were compiled (Table 1).

Table 1: Table 1. Datasets included in
nization.

Type	Organization	Name	Description
Aerial	CWS (NOR)	WLS1992	Scoter driven surveys in Labrador. Helicopter/fixed-wing and helicopter
Aerial	CWS (NOR)	WLS2008-09	Scoter driven surveys in Labrador. Helicopter/fixed-wing and helicopter
Aerial	CWS (NOR)	SES2017–19	Scoter driven surveys in northern NWT, Nunavut, northern MB and nor
Aerial	US FWS	WBPHS	Aerial transect surveys for waterfowl
Plot search	US FWS	FSMS	Duck populations, prairie potholes Area search
Point count	Birds Canada	BCMMP	BC Marsh Monitoring Programs
Point count	eBird	eBird	eBird stationary point counts less than 10 minutes
Point count	BU	DU_SFMN	Wetland bird point counts
Plot search	CWS	PRISM - PRA	Quadrat shorebird surveys
Point count	ECCC	YRBWS	Yukon Co-operative Roadside Breeding Waterfowl/Waterbird Survey (19
Aerial	ECCC	RLWS	Distribution and Abundance of Waterfowl in the Rasmussen Lowlands, N

Harmonization to WildTrax

The selected datasets were formatted to the format of the WildTrax "main" report for harmonization with other existing datasets. Datasets of ARU or point count style surveys were uploaded to the ARU and point count sensors in WildTrax, respectively, except for the eBird dataset, which is too large for storage in WildTrax. Aerial and plot search surveys were harmonized to the same data schema; however, were stored as .Rdata objects instead of upload to the WildTrax platform to avoid data misinterpreation because there are no appropriate sensors available for those datasets. Those harmonized datasets are available for download from a Google shared drive at https://drive.google.com/drive/folders/1n7lU7vGcl94dbL6ctlcJ1VKwGl9i6b3L. Code produced for data harmonization is available in a public Github repository at https://github.com/borealbirds/WildTrax-dataHarmonization.

Harmonization

The new datasets were then harmonized with the "BAM Dataset" (Barker et al. 2015), a collection of over 1.5 million bird surveys from across Canada to produce a comprehensive dataset of bird surveys. Following the definition of waterbirds as waterfowl, shorebirds, and marsh birds (i.e., excluding colonial waterbirds), we filtered the dataset to 142 species in 10 bird familes (Table 2): Anatidae (ducks and geese), Scolopacidae (sandpipers), Charadriidae (plovers), Recurvirostridae (stilts and avocets), Haematopodidae (osytercatches), Phalarapodidae (phalaropes), Rallidae (rails and coots), Ardeidae (herons and bitterns), Podicipedidae (grebes), and Gaviidae (loons).

The resultant dataset comprised 4.67 million surveys: 1.62 M from the BAM dataset, 1.00 M from eBird, and 2.05 M new surveys added through the data compilation and harmonization process, mostly from aerial transect and plot search surveys. The total number of waterbird observations across those surveys was 2.90 M. There were less observations than surveys because the database retained upland surveys where waterbirds were likely undetected; we retained these surveys in the database because they remain important and useful for understanding waterbird habitat relationships and species distribution modelling.

UPDATE THESE NUMBERS

Table 2: Table 2. Percentage of observations in harmonized dataset
by waterbird family and survey type.

Family	ARU	Aerial.transect	Plot.search	Point.count
Anatidae	1.06	39.73	16.23	19.82
Ardeidae	0.16	0.00	0.00	1.95
Charadriidae	0.09	0.00	0.00	4.14
Gaviidae	0.51	0.03	0.00	1.61
Haematopodidae	0.00	0.00	0.00	0.03
Podicipedidae	0.23	0.01	0.00	0.96
Rallidae	0.63	1.61	0.00	1.36
Recurvirostridae	0.01	0.00	0.00	0.12
Scolopacidae	2.47	0.02	0.00	7.21

Habitat data extraction

For habitat data extraction and subsequent analytical objectives, we filtered the harmonized dataset to just point-level surveys (i.e., ARU and point count surveys; 2.65 M surveys), for which habitat data can be readily extracted and compared across datasets.

We extracted several data sources of wetland habitat for each unique location in the point survey dataset using Google Earth Engine: - JRC Global Surface Water Version 1.4 (Pekel et al. 2016; 30 m resolution) was extracted as the mean within a 2 km buffer and the mean within a 20 km buffer – occurrence: the frequency with which water was present – seasonality: the mean number of months water was present – recurrence: the frequency with which water returns from year to year between 1984 and 2022 - Global Lakes and Wetlands Datavase Version 2 (Lehner et al. 2024; 15 arc second resolution) was extracted as the point value – dryland: the percentage that is classified as non-wetland – wetlandclass: the dominant wetland class - C-CAP Wetland Pontential (30 m resolution) was extracted as the mean within a 2 km buffer and the mean within a 20 km buffer – wetness: likelihood of wetness

Data Cleaning

The following section describes methods and results for data cleaning (Deliverables 3, 4 and 6).

Unknown Identification Treatment

Number of Unkwowns

First, we compared the frequency of unknown detections between ARU and in-person point count surveys to understand the magnitude of detections that are unknown waterbirds in acoustic recordings. There were approximately 25 times as many unknowns in ARU data as in point count data; however, this may be because unknowns are not consistently reported in point count protocols and/or go ignored by in-person observers due to the inability to pause the survey, as compared with an ARU observer. The majority of unknowns in both survey types were ducks and yellowlegs (Table 3). Unidentified scaup were also common in point count surveys, but not in ARU surveys because this pair of species is almost entirely nonvocal and therefore not well detected by ARUs. Unidentified shorebirds were also common in ARU surveys.

```
## Percent of unknowns in point count data:  0.014 %
```

```
## Percent of unknowns in ARU data:  0.359 %
```

```r
unidsp <- unid |>
  group_by(sensor, species_common_name) |>
  summarize(dets = n()) |>
  ungroup() |>
  mutate(Percent = ifelse(sensor=="ARU",
                          dets/(nrow(filter(unid, sensor=="ARU")))*100,
                          dets/(nrow(filter(unid, sensor=="PointCount")))*100),
         Percent = round(Percent, 2)) |>
  dplyr::select(species_common_name, sensor, Percent) |>
  pivot_wider(names_from=sensor, values_from = Percent, values_fill = 0) |>
  rename(`Unknown label` = species_common_name)
```

```
## `summarise()` has grouped output by 'sensor'. You can override using the
## `.groups` argument.
```

```r
kable(unidsp, caption = "Table 3. Percentage of unknown observations in point count and ARU surveys by u
  row_spec(0, bold = TRUE) |>
  kable_styling(full_width=FALSE)
```

Table 3: Table 3. Percentage of unknown observations in point count and ARU surveys by unknown label.

Unknown label	ARU	PointCount
Mallard / American Black Duck	1.77	0.00
Unidentified Common/Red-breasted Merganser	0.01	0.07
Unidentified Dabbling Duck	1.17	0.07
Unidentified Diving Duck	0.35	0.13
Unidentified Duck	47.04	64.38
Unidentified Goldeneye	0.28	3.03
Unidentified Goose	0.56	0.73
Unidentified Greater / Lesser Scaup	0.07	9.23
Unidentified Grebe	0.01	0.40
Unidentified Loon	1.36	0.79
Unidentified Shorebird	15.31	2.90
Unidentified Swan	0.04	0.73
Unidentified Teal	0.01	0.00
Unidentified Waterfowl	3.20	0.00
Unidentified Western/Clark's Grebe	0.01	0.00
Unidentified Yellowlegs	20.09	16.23
Unidentified quack	8.72	0.73
Unidentified Dowitcher	0.00	0.07
Unidentified Scoter	0.00	0.40
Unidentified Short-billed / Long-billed Dowitcher	0.00	0.13

Treatment of Unknowns

We then downloaded the "AI" report for HawkEars from WildTrax for as many recordings with unknown waterbird detections as available in the WildTrax system. HawkEars is a high-performance deep-learning classifier for Canada birds (Huus et al. 2025) and is run in WildTrax with a minimum score threshold of

0.3. For each unknown detection, we determined if there were any HawkEars detections in the same time window for a species in the potential group of candidate species. If so we replaced that unknown detection, with the HawkEars detection. If there were multiple species identified, we used the highest scoring one.

We downloaded the AI report for 89.6% of unknown identifications in ARU data. For those 89.6% unknown detections, we found matches with HawkEars for 26.9%. In other words, we reduced the proportion of unknowns in the overall dataset from 0.359% to 0.235%. The majority of replacements were of unidentified ducks and yellowlegs (Table 4); HawkEars was not very effective at replacing unidentified shorebird detections. The mean score of detections replaced by HawkEars was 0.69.

```
## Joining with `by = join_by(`Unknown label`)`
```

Table 4: Table 4. Percentage of unknown observations in ARU surveys by unknown label before and after HawkEars replacement.

Unknown label	Before	After	Mean score
Mallard / American Black Duck	1.77	0.88	0.56
Unidentified Common/Red-breasted Merganser	0.01	0.01	NaN
Unidentified Dabbling Duck	1.17	0.11	0.96
Unidentified Diving Duck	0.35	0.03	0.49
Unidentified Duck	47.04	27.30	0.61
Unidentified Goldeneye	0.28	0.10	0.73
Unidentified Goose	0.56	0.33	0.71
Unidentified Greater / Lesser Scaup	0.07	0.07	0.60
Unidentified Loon	1.36	1.12	NaN
Unidentified Shorebird	15.31	14.15	0.50
Unidentified Swan	0.04	0.01	NaN
Unidentified Waterfowl	3.20	2.64	0.66
Unidentified Yellowlegs	20.09	10.93	0.63
Unidentified quack	8.72	7.87	0.65

Upland Survey Flyover Removal

Instead of removing all upland surveys from the dataset, which would remove data points useful for applications like understanding habitat relationships, we completed two separate tasks: 1. Included habitat information that can be used to filter the dataset to wetland observations only (see 'Habitat data extraction' above); 2. Removed waterbird observations from the dataset that were at upland locations and had a high probability of being a flyover observation. In other words, an observation of an individual or group of waterbirds that were flying over the survey area and were not actually using the habitat for nesting or foraging purposes.

We removed high probability flyover observations from the dataset by modeling the probability that an observation was a flyover relative to our extracted habitat covariates from a subset of the point count surveys in the dataset that included and differentiated flyover observations as part of the survey protocol. There were just over 40 thousand observations in the dataset used for modelling flyover probability across nine of the ten waterbird families included in the dataset (Table 5). The total frequency of flyover observations in that dataset was 4.09%.

```
## `summarise()` has grouped output by 'Family'. You can override using the
## `.groups` argument.
```

Table 5: Table 5. Percentage of observations in point count surveys that were flyovers by waterbird family.

Family	Percent flyovers
Anatidae	5.108521
Ardeidae	12.384259
Charadriidae	1.471810
Gaviidae	6.976744
Podicipedidae	10.461538
Rallidae	6.318957
Recurvirostridae	2.072539
Scolopacidae	4.097059

We explored several modelling approaches, including logistic regression with and without random intercepts for waterbird family (GLM, GLMM), generalized additive modeling with a random intercept for family (GAM), and boosted regression trees with a fixed effect for family (BRT). We withheld ten percent of the dataset and compared the predictive performance of each modeling approach using the area under the curve of the receiver operating characteristic (ROC AUC) and the log loss likelihood (Likelihood) of the model, both for the training and the testing dataset (Table 6). Overall, the boosted regression tree showed some signs of overfitting via lower performance on the test dataset than the training dataset; however, the performance on the test dataset was still better than any of the other three model types.

Table 6: Table 6. Predictive performance of four models for probability of waterbird flyover on training and testing datasets.

Model	Dataset	AUC ROC	Likelihood
BRT	test	0.8787819	0.1199498
BRT	train	0.9575960	0.0904343
GAM	test	0.8353738	0.1418539
GAM	train	0.8276407	0.1403627
GLM	test	0.7455761	0.1575209
GLM	train	0.7437021	0.1530956
GLMM	test	0.7707342	0.1530936
GLMM	train	0.7715277	0.1488297

The wetland covariates extracted with the larger, 20 km radius buffer were the explained the most variation in the boosted regression tree model, particularly the percent dryland variable (18.7% variation explained), with wetland class and waterbird family also explaining 10.7% and 7.4% respectively. The covariates extracted at the 2 km radius each explained about 4% of the total variation explained by the model.

We made predictions for the probability of flyover for each detection in the overall survey dataset using the boosted regression tree model. There were no observations of Haematopodidae (oystercatchers) in the flyover training dataset, so we set observations for that waterbird family to Scolopacidae (sandpipers). We used a threshold of 20% probability of detection to label detections as flyovers for several reasons: 1. There was a break in the distribution of probability estimates at 0.20; 2. A value of 0.20 maximized the precision and recall of the model; 3. Thresholding with 0.20 identified approximately 7% of the detections in the dataset as flyovers, which was similar to the proportion in the flyover training dataset. We removed all waterbird observations over this threshold from the waterbird dataset.

Data Analysis

The following section describes methods and results for data analysis (Deliverable 5), including additional exploration of waterbird detection frequency.

Wetland Habitat Associations

We used the entire harmonized and cleaned point survey dataset to summarize habitat relationships for each species with the extracted habitat data. We were interested in a high-level summary of habitat relationships, not density per se, and so we converted the count information in the waterbird dataset to presence/absence and used logistic regression for habitat modelling. We modelled the probability that a species was reported on a given survey relative to the habitat covariates as well as second order polynomials for the hour of day and day of year. For each model, we randomly selected one observation from each location to avoid pseudoreplication.

Detection Occurrence Summary

In addition to the initial objectives and deliverables in the statement of work, we reported on one additional component of the waterbird database: detection occurrence. In particular, we were interested in informing which species are detected at a similar occurrence rate with ARU surveys as point count surveys. For each species in the dataset, we calculated the following metrics for ARU surveys and for point count surveys: -Survey_occurrence: the total proportion of surveys during which the species was detected; -Site_occurrence: the proportion of sites at which the species was detected during at least one survey; -Detection_rate: the proportion of surveys during which the species was detected for sites where it was detected during at least one survey; -Occurrence_count: the mean count of the species when detected during a survey.

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